

# Kasra Farsoudi

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## EDUCATION

### University of Southern California

Los Angeles, CA

*B.S. in Computer Engineering & Computer Science*

*Aug. 2024 – Dec. 2026*

**Relevant Coursework:** Professional C++, Machine Learning, MOS VLSI Circuit Design, Parallel Design, Data Structures, Embedded Systems, Digital Circuits, Distributed IoT Systems, Algorithms & Theory of Computing, Software Development, Operating Systems, Computer Architecture, Networking

## EXPERIENCE

### Full-Stack Software Engineer

Jul. 2023 – Present

*Brake Performance*

*Los Angeles, CA*

- Implemented Apple Pay end-to-end on an e-commerce platform with frontend payment session handling on Next.js, and backend charge processing with PHP Symfony, increased revenue by 20%.
- Devised a fraud-detection engine using MySQL that indexes new orders against flagged orders.
- Automated a cron for AWS Linux that ingests shipment data from UPS & USPS and updates order status.
- Engineered a full-stack sales report system with an SQL data ingestion pipeline and React Admin, providing day/month/year revenue breakdowns by order, part, and product.
- Established a reusable transactional email system for wholesale approvals, shipment errors, and fraud alerts

### Junior Software Engineer (*Lead Frontend Engineer*)

October 2022 – July 2023

*I-Safe*

*Carlsbad, CA*

- Led the frontend team through agile sprint planning, mentoring new interns and delegating work across the group.
- Developed the front-end logic for handling 25+ Rest API endpoints in TypeScript on React Native.
- Built the Redux state management system, including user data storage and synchronization to React lifecycle.
- Streamlined Redux-Saga middleware for RESTful API calls & pipelining asynchronous data to client-side storage.
- Spearheaded the entire login and authorization workflow, from design to implementation.
- Integrated secure token-based authentication with refresh logic in JavaScript.

## PROJECTS

### Self-Hosted AI Agent | *Ollama, Fedora Linux, Docker, PostgreSQL, RAG, n8n*

August 2026

- Architected a fully private AI assistant stack on a headless RTX 3090 Fedora server, accessed over Tailscale
- Deployed local LLM inference with Ollama, serving Qwen 3.8 27B
- Configured a LibreChat (Docker) chat UI wired to Ollama endpoint, with custom MCP tooling
- Unified MongoDB, Meilisearch, and pgvector backends for conversations, search, and RAG
- Designed a cross-device persistent memory service in FastAPI + SQLite
- Orchestrated an n8n + Postgres automation layer: an LLM email agent classifying Gmail and extracting deadlines, writing approved items to Google Calendar/Tasks

### Paid LLM Inference Gateway | *Go, x402, PostgreSQL, Ollama*

September 2026

- Shipped a Go gateway that fronts a self-hosted Ollama instance with metered, pay-as-you-go inference
- Funded prepaid USDC balances via the x402 HTTP 402 payment protocol, integrating Coinbase's facilitator on Base to verify and settle on-chain payments
- Modeled an off-chain Postgres ledger to credit balances and record auditable usage transactions

### FPGA Street Fighter | *Verilog, Xilinx Vivado*

April 2025

- Synthesized a 2-player fighting game on Nexys A7 FPGA hardware with custom hard-wired breadboard controllers
- Programmed the core game engine in Verilog: collision detection, combat resolution, and win-condition logic
- Built the player state machine and a reusable timer module driving movement, jump arcs, and shield mechanics

## TECHNICAL SKILLS

**Languages:** Go, C/C++, JavaScript, SQL (MySQL & Postgres), PHP, Verilog, CUDA, Python

**Frameworks:** React / React Native, Node.js, Symfony, API Platform, ESP32 / ESP-IDF, MongoDB

**Developer Tools:** Git, Docker, Tailscale, n8n, Neovim, Ollama, AWS

**Linux Distros:** Arch (Personal Distro), Amazon Linux, CentOS, Fedora, Ubuntu, Raspbian